UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

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Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This paper presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este artículo se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

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Chapter 1

Introduction

1.1 Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5% of energy sector CO_2 emmisions [1] [2], has continuosly increased in emissions, as shown in figure 1.1.

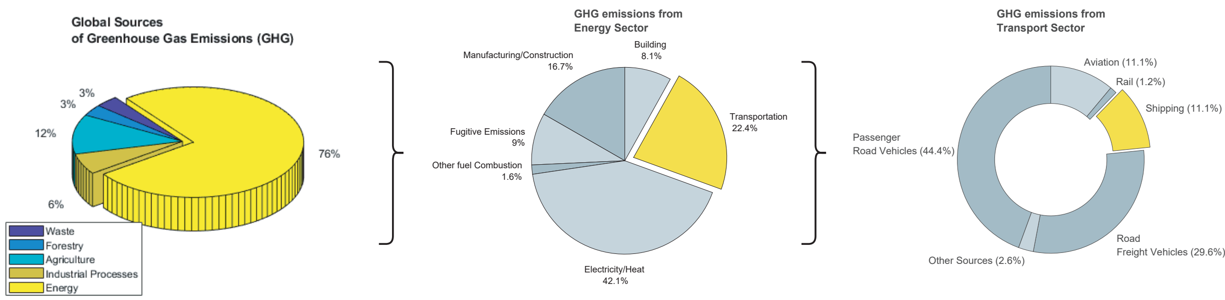
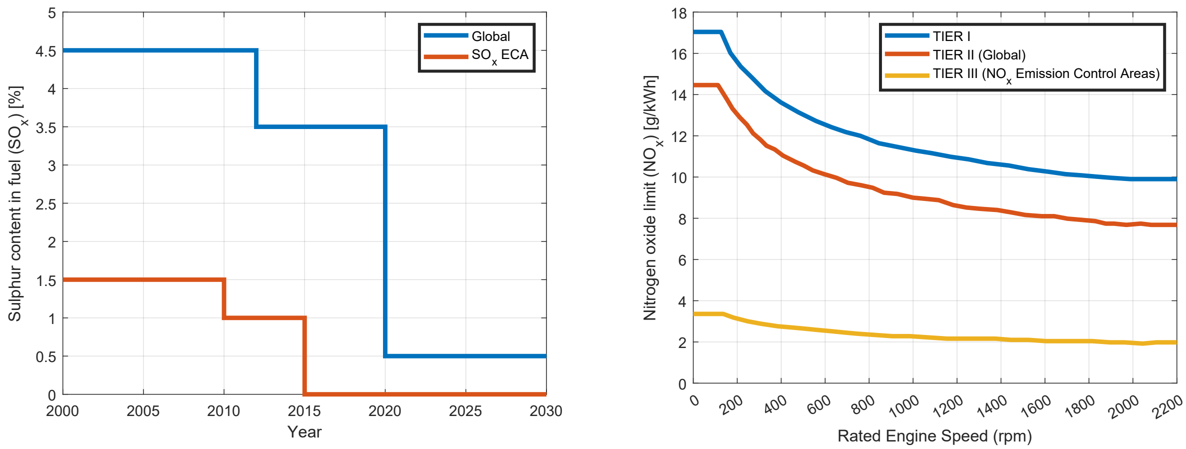


Figure 1.1: Global sources of greenhouse gas emissions and breakdown of CO_2 emissions from the energy sector and the transport subsector [1].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and provinding incentives. The initial strategy avoids a reduction in total GHG emissions from international shipping by at least 50% by 2030 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5% mass by mass (m/m) by 2025 per the International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI, as shown in figure 1.2a. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed, as shown in figure 1.2b. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.



(a) SO_x limits. (b) NO_x limits.

Figure 1.2: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs perspective).

To comply with the standards and emissions reduction goals set by IMO, the maritime industry is looking for reduced and zero-emission technologies and approaches. The transition from traditional fossil fuel engines to zero-emissions technologies is challenging in a competitive market, where manufacturers are struggling to increase revenue and reduce production cost. The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

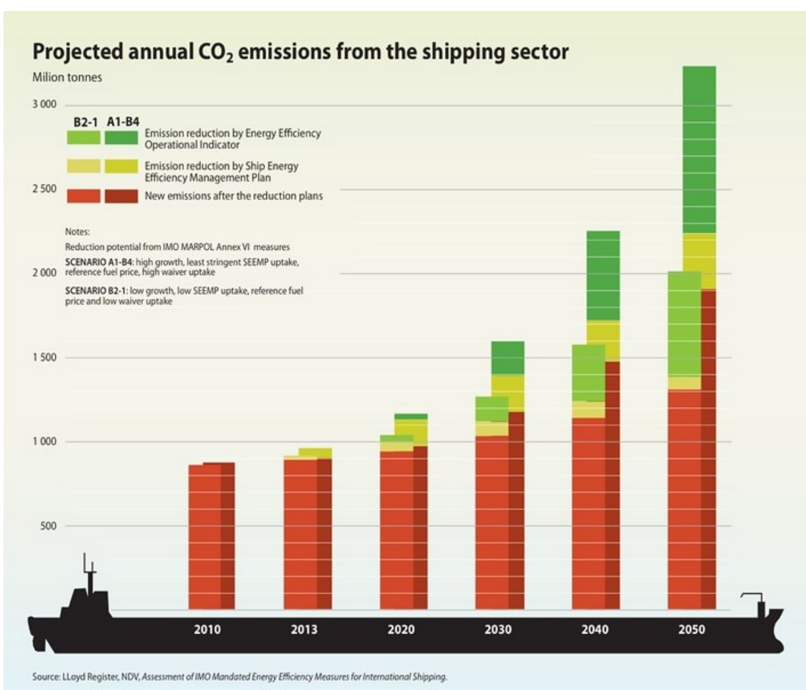


Figure 1.3: IMO greenhouse gas emissions strategy [6].

1.1.1 Propulsion Architectures

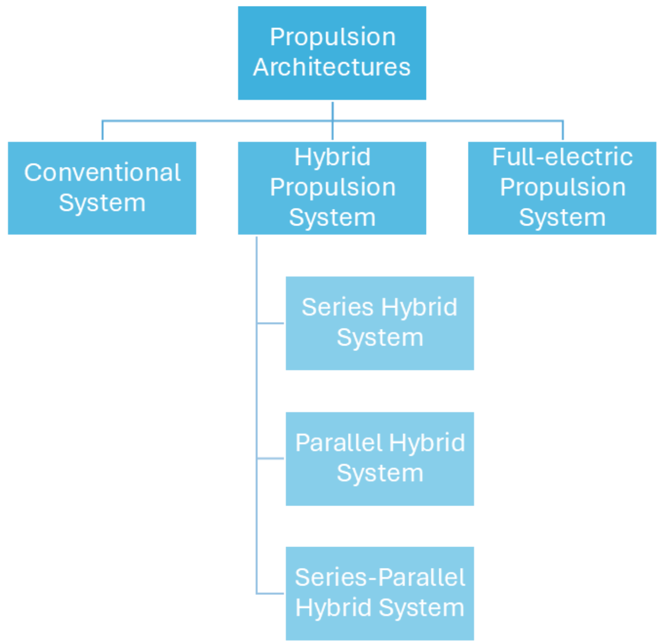


Figure 1.4: Types of propulsion systems−− > PDF

While conventional maritime transport is marked by the low technological development of the shipping and maritime industry, especially with regard to propulsion systems, which translates into a continuous increase in fossil fuel consumption, with the environmental impact that this represents, fully electric and hybrid propulsion architectures, made possible by innovative and high-performance energy storage systems (batteries, fuel cells, supercapacitors, etc.), are a promising way to significantly reduce fossil fuel consumption, pollutants, and greenhouse gas emissions [4].

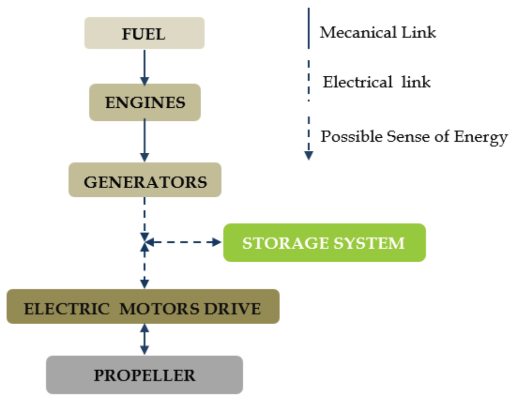
New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [5].

For full-electric propulsion, the main advantages include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller.

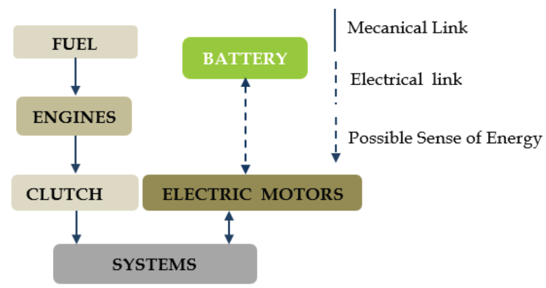
The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [5]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significicatly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [4].

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different power sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

As can be seen in the following subsections, there are three main architectures for hybrid propulsion chains [4]:



(a) Series Hybrid architecture [4].



(b) Parallel Hybrid architecture [4].

Figure 1.5: Semi-hybridization.

a) Series Hybrid Architectures

In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter), as can be seen in the figure 1.5a. The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electric power or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electric power is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

b) Parallel Hybrid Architectures

Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches, as can be seen in the figure 1.5b. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel

Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

c) Series-Parallel Hybrid Architectures

The series-parallel architecture combines both architectures defined previously. This association allows the passage of a parallel-type operation to a serial type operation.

Hybrid Propulsion systems are based on the combination of several power sources and energy storage systems.

1.1.2 Power Generation Systems

a) Steam Turbine

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it to do mechanical work on a rotating output shaft [10] [11]. Steam turbines work by expanding steam through a series of blades mounted on a shaft, which turns a propeller to move the ship. They are particularly favoured for their smooth operation, high power output, and efficiency at high speeds, making them suitable for large vessels like liquefied natural gas (LNG) carriers and naval ships.

Steam turbines revolutionized ship propulsion with their introduction in the late 19th century, offering advantages such as small size, reduced maintenance, and low vibration. Despite the need for precise reduction gears or turbo-electric transmission to match the high RPM of turbines to the effective propeller speeds, the benefits included light weight and compact engine rooms.

b) Gas Turbine

Gas turbine engines are defined as aerospace engines that convert the chemical energy in fuel into thermal energy through combustion with air, resulting in the rapid expansion of gas that can perform work in the turbine and exhaust nozzle [12].

The gas turbine engine is recognized for its relative mechanical simplicity and its capacity to use a broad spectrum of fuels. The excellent power–weight ratio of gas turbine engines can be effectively exploited. The performance of gas turbine engines is affected by compressor efficiency, turbine efficiency, pressure ratio, peak cycle temperature, and combustion efficiency. Compressor and turbine efficiencies are continuously increasing through better design and manufacturing techniques, but little room is left for improvement. Increasing the pressure ratio can significantly improve the efficiency of an engine, but in order to retain sufficient power output, an increase in pressure ratio must be accompanied by an increase in maximum cycle temperature.

The first time a gas turbine was mounted on a watercraft occurred in 1947 in the United Kingdom, when a diesel free power turbine was mounted on a modified gunboat as a proof of concept [13]. This craft endured sea trials for 4 years, until the technology was proven to be a viable concept for the future of naval propulsion. HMS Grey Goose became the first fully gas turbine powered ship, being launched in the UK in 1953. It served in the British Navy for only four years; however, it has fulfilled its mission of proving the viability of gas turbine propulsion at sea. Since the inception of this technology, gas turbine engines or combinations of them with other engine technologies have been utilized in a wide range of watercraft, ranging from hovercraft all the way up to full-size aircraft carriers, with power outputs starting at single digits of MW all the way to hundreds of MW.

c) Diesel Engines

A diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [4]. Diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [7]. Compared to other prime movers such as a gas turbine or steam turbine, the diesel engine has a high efficiency. Its low fuel consumption is the main reason it is widely used in marine applications.

This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power.

1.1.3 Energy Storage Systems

a) Supercapacitors

Supercapacitors can be interesting intermediate energy storage systems for small ships with short power transients [4]. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with slower systems, such as batteries and fuel cells, to manage high power short transients. They can operate at very low temperatures at which many types of electrochemical batteries stop working.

b) Flywheels

Flywheels can be used in some particular cases. Like supercapacitors, they are characterized by a high power density but a relatively low energy density. However using flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

c) Batteries

The oldest and most classical method for storing electrical energy is using batteries. Battery technology has matured significantly during the past two decades as not only the world but also energy has become increasingly mobile [19].

Table 1.1: Energy Storage Systems Comparison [20].

Technology	Battery	Supercapacitor	Flywheel
Energy Density (Wh/kg)	50-250	0.05-5	5-100
Power Density (W/Kg)	50-2000	100k	1000
Lifecycle (cycle)	400-9000	50k	20k-100k
Lifetime (years)	5-10	5-8	15-20
Overall Efficiency (%)	85-99	60-65	15-20

As shown in the table 1.1, the battery provides the highest power density but lacks overall efficiency compared to energy storage systems. The flywheel, on the other hand, is the most durable system, offering a cycle life of between 20,000 and 100,000 cycles and a service life of between 15 and 20 years. However, due to its low power density, it may not provide enough power for a vessel.

The above comparison helps to understand why batteries are usually chosen over other types of energy storage systems: because of their reliability and high energy density.

1.1.4 Battery Systems in Maritime Applications

In 2018, the European Maritime Safety Agency (EMSA) commissioned DNV GL to perform a study on the use of electrical storage systems in shipping, with the objective of providing an overview of technology, research, feasibility, regulations and safety of battery systems in maritime applications [18].

Batttery application onboard ships can have multiple functional roles. While batteries can fully power a vessel for short distance or duration, improving performance and energy efficiency of the overall vessel is often the key purpose. Among the different uses of batteries on ships are:

- Spinning reserve
  - Backup for running generators.
  - Fewer generators needed online.
- Peak shaving
  - Act as a buffer.
  - Level power seen by engines.
- Optimise load
  - Optimise the operating point of the generators.
  - Reduce maintenance.
- Immediate power
  - Instant power in support of generators.
- Harvest energy
  - Recover energy from cranes, drilling equipment, etc.
  - Accommodate energy from renewables.
- Backup power
  - Battery system provides backup power, UPS like functionality.

Table 1.2: Energy Storage Systems Comparison [20]

Battery	Energy Density (Wh/kg)	Vol. Energy Density (Wh/L)	Cost (USD/kWh)	Lifecycle	Efficiency (%)
Lead-Acid	20-40	60-110	165	500-1000	50-95
Ni-Cd	20-60	60-150	-	2000-2500	70-90
Ni-MH	50-70	140-300	146	500-2000	89-92
Na-S	~120	150-300	400	>1500	86-92
Li-ion	50-250	250-675	200-600	400-9000	85-99
Li-polymer	~150	250-730	-	-	-
Li-S	400	350	400-500	1500	-

Different ships have different operating profiles and batteries must respond to specific energy and power demand while also having in consideration the desired/expected life-cycle for the battery [18].

Considering the incorporation of batteries into the propulsion system of vessels, different propulsion architectures are recognized that provide a different use for batteries, even without considering them, as is the case with conventional propulsion architecture [18] [19]. This can be seen in the figures 1.6, 1.7 and 1.8.

a) Diesel-mechanic engine propulsion

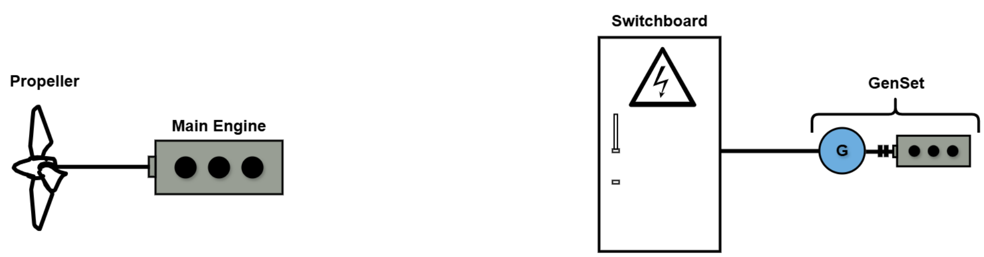
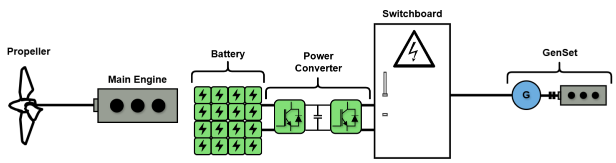


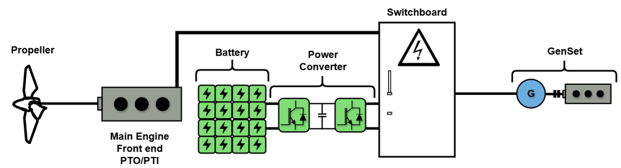
Figure 1.6: Traditional diesel-mechanic propulsion of a large merchant vessel −− > PDF.

In general, the configuration shown in Figure 1.6 is the most common installation on large ocean-going vessels [19]. It consists of a two-stroke main engine responsible for propulsion, while the ship's load is covered by auxiliary engines/sensets installed on board the vessel.

b) Diesel-mechanic engine hybrid propulsion



(a) Semi-hybrid diesel-mechanic propulsion of a large merchant vessel with hybrid electric grid −− > PDF



(b) Full-hybrid diesel-mechanic propulsion of a large merchant vessel with front end PTO/PTI from RENK −− > PDF.

Figure 1.7: Semi-hybridization.

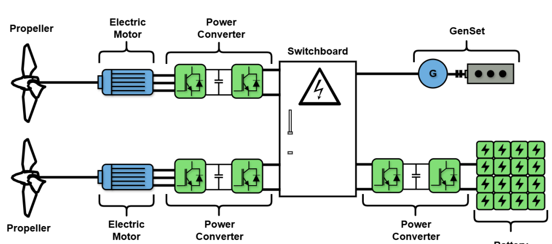
In Fig. 1.7a, a battery is included in a traditional system where the electric grid is separated from the propulsion of the vessel [19]. The battery will in such a solution be effective for smoothing the connected hotel electrical loads and contribute to handle large load steps or peaks [18]. When large load steps are increased, the number of auxiliary motors also increases. In some cases, the load can regenerate energy, for example, in crane operations. In such cases, the battery can be used to capture the energy.

In 1.7b, a battery is included in combination with a shaft generator power take-out (PTO) and power take-in (PTI) [19]:

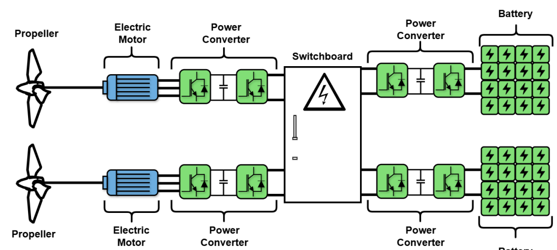
- **Power Take-Out (PTO):** The electrical/mechanical hybrid solution allows electricity to be generated by the main engine.
- **Power Take-In (PTI):** The electrical/mechanical hybrid solution allows propulsion power to be produced by generator sets and batteries. Additional power is possible (boost mode) when the main engine and PTI motor are running in parallel.

In addition, a ship with a hybrid electric/mechanical propulsion system may have a direct current distribution system, which can adjust the speed of the primary generator engines to the optimum fuel level depending on the load. This will reduce fuel consumption and minimize environmental impact [18].

c) Hybrid/electric propulsion with electric motors



(a) Full-hybrid diesel-electric propulsion with batteries in a hybrid system −− > PDF



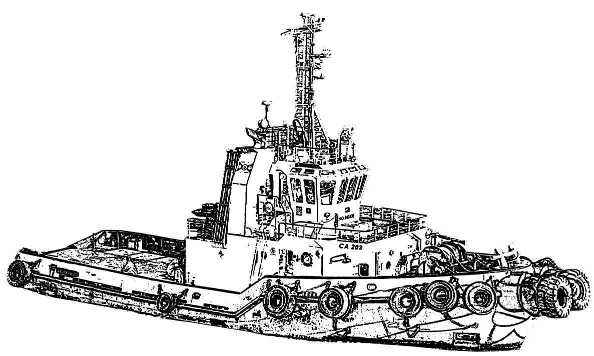
(b) Pure battery-electric propulsion −− > PDF.

Figure 1.8: Electric motor incorporation.

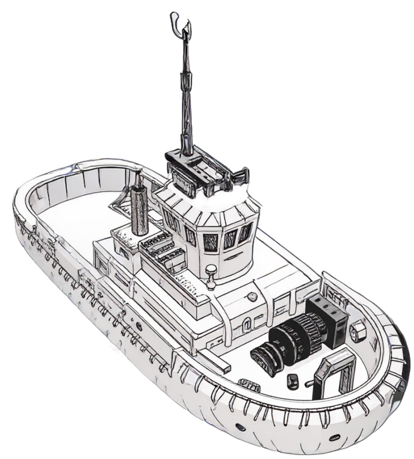
A full-hybrid diesel-electric solution, as is shown in fig. 1.8a, reduces the noise and vibration level on the ship [18]. Supplementary to be a source of energy for propulsion, the batteries will also smooth the load variations on the generator sets. This type of solution can for instance facilitate the use of zero emission operation when entering a harbour. Another version of this solution is with the batteries distributed into the drives directly [18]. In this way, each propulsion unit is independent of a common power source, which could be valuable for vessels that require highly reliable propulsion thrust.

In the fig. 1.8b, the propellers are connected to electric motors, which are driven by the energy stored in a battery system that is typically charged from shore [19]. In some battery-electric systems, a smaller diesel generator is sometimes included to ensure operation if the batteries fail to charge or to enable longer voyages, i.e. when the vessel has to dock [19]. For such a solution the batteries will be charged through an AC/DC converter. The converter can either be located on board the vessel or on shore. The fig. 1.8b shows two independent battery systems that deliver power to the thruster. This is according to class rules, which require that two independent battery systems are installed to provide propulsion power in case one of the systems fail [18].

1.1.5 Tugboats



(a) Conventional tug representation.



(b) Azimuth Tug representation

Figure 1.9: Tugs representations

Tugboats are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [20]. Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [3].

A tug, or more commonly a tugboat, is an assistance boat which helps in the mooring or berthing operation of a ship by either towing or pushing a vessel towards the port [9] [14].

A tug is a particular class of boat which helps mega-ships enter or leave a port. In addition to towing the vessel towards the harbour, tug boats can also be engaged to provide essentials, such as water, air, etc., to a vessel.

Tug boats are also essential for non-self-propelled barges, oil platforms, log rafts, etc. Due to their solid structural engineering, tugs are small but relatively powerful machines.

The usage and functions of tugs vary from port to port, as different ports have different requirements and intakes. The common thing is pushing or towing mega boats or barges. Their usage depends on the following factors:

- Port traffic Volume
- Types of ships to be served by that tug
- Navigational obstacles to be catered to
- Conditions of environmental protection
- Local laws and
- Domains to be carried by a tug

An average tug boat has 500-2500 kW engines (680-3400 hp), but those which are larger and venture out into deep waters have engines with a power close to 20000 kW (27200 hp) and a power/tonnage ratio ranging between 2.20-4.50 for large tugs and 4.0-9.5 for harbour tugs. These are high ratios, considering the ratio of cargo ships or vessels varies between 0.35 and 1.20.

Based on purpose, they serve marine tugs can be of two types:

- Escort Tugs: The tugs designed generally to escort and manouvre ferries and barges to their destination are known as escort tugs.
- Support Tugs: These tugs provide support services for offshore and towing operations. These tugs play a significant role in berthing operations.

Mainly, tugs used in the marine industry are of three types, which are listed:

1. Conventional tug
2. Tractor tug
3. Azimuth tug

The efficiency of these types of vessels is based on fuel consumption in relation to load capacities, always taking into account both the propulsion system and the energy conversion system that determines it. The efficiency of conventional propulsion systems varies between 38% and 41% for a power range between 100 kW and 2,500 kW. For vessels with higher power ratings, efficiency increases to 48% [15] [16]. Low efficiency is further affected by the variability of the operating and load conditions to which engines are subjected. This variability leads to an increase in specific fuel consumption, which causes higher emissions and energy waste through exhaust gases and cooling systems [15]. However, this level of efficiency could be optimized by the hybridization.

1.2 Challenges and Research Opportunities

Several studies have examined hybrid propulsion systems [8]. A genetic algorithm to obtain the optimal operating points regard with series hybrid configurations in power generation, especially for the case of heavy duty hybrid drive applications [17]. A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [21]. The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [22]. A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and lower CO2 emissions [23].
However, there is no clear and detailed methodology for the hybridization of vessels such as tugboats that would provide a path from the conceptual stage to the technical details and subsequent implementation.

1.3 Thesis Objectives and Outline

The general objective of this work is to develop a methodology for the design and sizing of the battery bank for a hybrid tugboat, which will allow us to determine the benefits that would be obtained from hybridization.

Specific objectives include:

- Model the operation of a conventional tugboat
- Define hybrid architecture.
- Propose the service power of the electrical system.
- Preliminary design of the battery bank based on technical and economic criteria.
- Simulation of the proposed cell and battery bank.

1.4 Methodology

The methodology to follow for the development of this work is:

- **State of Art:** Review of existing hybrid propulsion system types and battery technologies.
- **Mathematical Analysis and Modeling:** Theoretical study of the tugboat's load profile and the implementation of batteries to support the vessel's operation.
- **Results:** Simulation of the models obtained to complete the conceptual analysis.

1.5 Scope and Limitations

1.6 Chapter Summary

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